GeMS: Geometry-aware Mesh Splatting

Shao Jie Hu Chen
TUM
shaojie.hu@tum.de

Pablo Díaz Viñambres
TUM
pablo.diaz@tum.de

Abstract

*In recent years, Gaussian Splatting has dramatically improved the quality of Novel View Synthesis (NVS) pipelines, while also allowing for geometry extraction via a separate postprocessing step. A recent work, MeshSplatting, skips this step by jointly optimizing geometry and appearance via opaque triangle primitives. However, extracted meshes are often noisy and tend to drift off indoor structural components. In this work, we propose **GeMS**, a method which incorporates planar decompositions and geometric regularizers. We also enforce semantic separation for reducing triangles shared between multiple object instances. Our experiments on the Mip-NeRF360 and ScanNet++ datasets show that GeMS improves the geometry of the generated meshes (especially in planar surfaces) and maintains crisp object boundaries.*

1. Introduction

Reconstructing dense, explicit 3D geometry of indoor scenes from posed RGB images is a core problem for downstream tasks such as robotics, AR/VR, and scene understanding [1, 8, 11, 12, 15]. Radiance-field and splatting-based methods [6, 7, 9, 10] have made photorealistic novel-view synthesis fast and high quality. However, their underlying representations (implicit fields, unstructured Gaussians) are not directly usable as a mesh, required by downstream applications.

In this work, we present **GeMS**, a geometry-aware extension of mesh-splatting that biases the reconstructed mesh toward the planar, object-structured geometry characteristic of indoor scenes. The key observation is that the original MeshSplatting struggles to correctly reconstruct planar structures overrepresented in indoor scenes, and obtained geometry can be conditioned by applying additional regularization during training. This is achieved by incorporating three additional regularization terms:

(M1) A Laplacian geometry and normal consistency term that damps high-frequency vertex noise and penalizes dihedral folding between adjacent faces.

(M2) A planar alignment term that embeds planar priors into the training scheme.

(M3) A semantic boundary term that discourages shared triangulations between different objects.

The remainder is organized as follows. Sec. 2 reviews related work on radiance-field and splatting-based reconstruction. Sec. 3 details the different regularization terms introduced in the training scheme. Sec. 4 describes our experimental setup, results, and ablations of each regularization term. Sec. 5 concludes and discusses limitations and future work.

2. Related Work

Gaussian Splatting 3D Gaussian Splatting [9] represents a scene as a set of explicit, rasterizable Gaussians, enabling real-time novel-view synthesis. However, it does not yield a surface mesh directly. Closest to us, MeshSplatting [5] optimizes an explicit triangle mesh end-to-end against the photometric objective, but the obtained geometry is broken, especially in planar surfaces.

Geometric regularization of scenes Several works in the literature have constrained the training scheme to add geometric regularization. 2D Gaussian Splatting [7] flattens Gaussians onto surfels and supervises rendered depth and normal maps with distortion and consistency losses, while SuGaR [3] regularizes Gaussians to lie flat on and align with an underlying surface for cleaner mesh extraction. Triangle Splatting [6] introduces the differentiable triangle parameterization, and MeshSplatting [5] builds on it with a shared-vertex structure and restricted-Delaunay connectivity. These regularizers keep surfaces locally smooth and well-conditioned, but they are geometry-only and architecture-agnostic: none exploit the large planar surfaces or object structure that dominate indoor scenes.

Semantic-aware reconstruction Semantic information helps to alleviate inconsistent 3D reconstruction. Panoptic Lifting [13] lifts inconsistent 2D panoptic masks into a coherent 3D panoptic radiance field, resolving multi-view

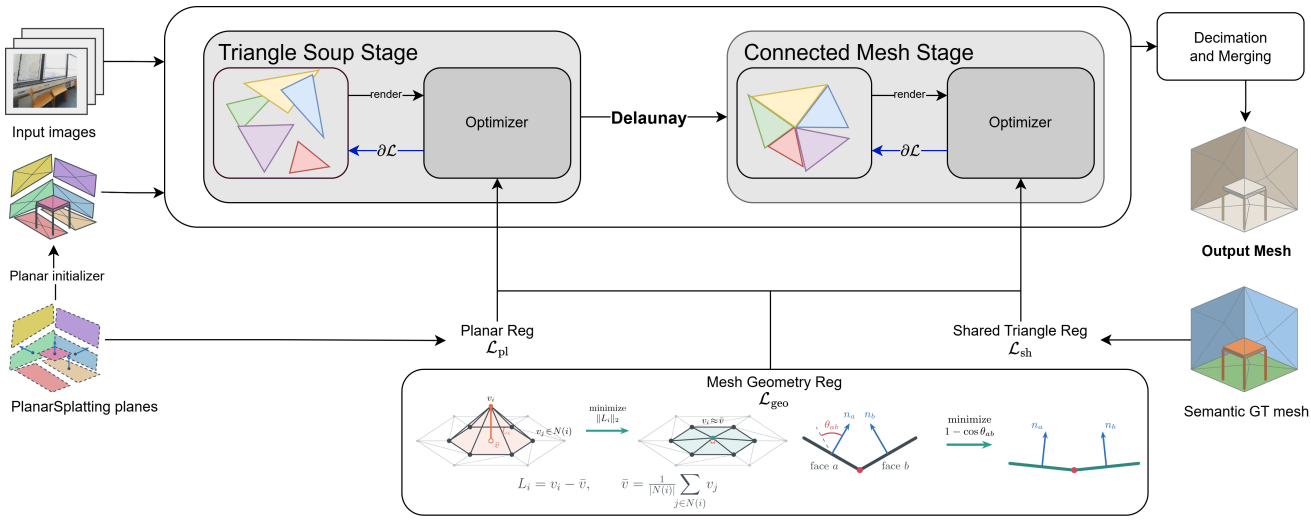


Figure 1. Overview of **GeMS**. Given posed RGB images of an indoor scene, we detect rectangular plane primitives with PlanarSplatting and merge them into area-weighted planes. Training is regularized via a local-smoothness loss $\mathcal{L}_{\text{geo}}$, a plane-alignment loss $\mathcal{L}_{\text{planar}}$ and a semantic separation loss $\mathcal{L}_{\text{shared}}$.

label conflicts during optimization. For splatting representations, Semantic Gaussians [4] distills 2D features from pretrained image encoders into a semantic component of each Gaussian for open-vocabulary scene understanding, and Gaussian Grouping [16] augments each Gaussian with an identity encoding supervised by foundational models. These methods attach semantic labels or features to an existing representation as a queryable overlay, but they leave the underlying geometry unchanged. In contrast, we use semantic labels as a geometric constraint on an explicit mesh.

3. GeMS

We present GeMS, a method which explicitly regularizes the mesh generated by MeshSplatting [5]. To do so, we add three new regularization terms in training, as summarized in Fig. 1.

3.1. Planar-guided initialization

In MeshSplatting, initialization consists of randomly oriented triangles from the SfM cloud points. In GeMS, we first run PlanarSplatting [14], obtaining a set of planar primitives p with unit normal $\hat{\mathbf{n}}_p \in \mathbb{R}^3$, center $\mathbf{c}_p \in \mathbb{R}^3$, and half-extents $(r_{x,p}, r_{y,p})$, giving it area

$$a_p = 4\, r_{x,p}\, r_{y,p}. \tag{1}$$

While these primitives are already useful, they cover small parts of the scene, making them noisier for initialization and regularization purposes. We thus merge these primitives into a plane m with constituent primitives P_m. The merged center and unit normal are the area-weighted mean center and the normalized area-weighted mean normal of P_m, and each merged plane is sampled into a grid of triangles proportional to its area $a_m = \sum_{p\in P_m} a_p$, concatenated with the (pruned) SfM-based triangle soup to form the initial mesh.

3.2. Mesh regularization losses

While planar initialization helps with achieving scene completion, the under-constrained photometric objective still generates meshes with degraded quality. We aim to fix this issue with geometry regularizers.

Local smoothness. For improving local smoothness, we add a smooth Laplacian term damping high-frequency vertex noise, and a normal-consistency term penalizing high dihedral angles between adjacent faces:

$$\mathcal{L}_{\text{lap}} = \frac{1}{|V|}\sum_{i\in V}\left\| v_i - \frac{1}{|N(i)|}\sum_{j\in N(i)} v_j \right\|_2, \tag{2}$$

$$\mathcal{L}_{\text{nc}} = \frac{1}{|E|}\sum_{(a,b)\in E}\big(1-\cos\theta_{ab}\big), \tag{3}$$

$$\mathcal{L}_{\text{geo}} = \mathcal{L}_{\text{lap}} + \mathcal{L}_{\text{nc}}, \tag{4}$$

where V is the vertex set, $N(i)$ is vertex i's 1-ring, E is the set of interior edges shared by two faces, and θ_{ab} is the angle between the normals of the two faces sharing an edge.

Plane alignment. For every triangle still assigned to a source plane with unit normal $\hat{\mathbf{n}}_t$ and centroid $\mathbf{c}_t$, we penalize both positional drift off the plane and orientation drift of the face normal $\hat{\mathbf{f}}_t$:

$$\mathcal{L}_{\text{dist}} = \frac{1}{|T_p|}\sum_{t\in T_p}\frac{1}{3}\sum_{k=0}^{2}\left|(v_{t,k}-\mathbf{c}_t)\cdot\hat{\mathbf{n}}_t\right|, \tag{5}$$

$$\mathcal{L}_{\text{normal}} = \frac{1}{|T_p|} \sum_{t \in T_p} \left(1 - |\hat{\mathbf{f}}_t \cdot \hat{\mathbf{n}}_t|\right), \tag{6}$$

$$\mathcal{L}_{\text{planar}} = \mathcal{L}_{\text{dist}} + \mathcal{L}_{\text{normal}} \tag{7}$$

where T_p is the set of triangles currently assigned to a plane and $v_{t,k}$ is the k-th vertex of triangle t. Since the Delaunay triangulation step that runs in the middle of the optimization process of MeshSplatting invalidates per-triangle plane assignments, we reassign each triangle to its nearest plane rectangle before continuing.

3.3. Semantic-boundary refinement

We want boundaries to fall between faces, not cut through them, so we both discourage straddling triangles from persisting and actively refine the mesh to resolve them.

We first transfer semantic labels from the ground truth to our current set of vertices via nearest-neighbor vertices. Then, we flag any triangle whose three corner labels disagree and softly penalize the rendered opacity of every vertex touching one,

$$\mathcal{L}_{\text{shared}} = \frac{1}{|V_\partial|} \sum_{i \in V_\partial} \alpha_i, \tag{8}$$

where V_∂ is the set of vertices touching at least one currently straddling triangle and α_i is vertex i's rendered opacity. Furthermore, a periodic subdivision pass bisects the mesh edges that cross a label boundary, so straddling triangles are resolved with the least extra geometry possible.

3.4. Training objective

All terms act on the same mesh and are optimized jointly with the photometric objective $\mathcal{L}_{\text{photo}}$ of MeshSplatting, a combination of an L_1 and a D-SSIM term between rendered and ground-truth views:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{photo}} &+ \lambda_1 \mathcal{L}_{\text{lap}} + \lambda_2 \mathcal{L}_{\text{nc}} \\ &+ \lambda_3 \mathcal{L}_{\text{dist}} + \lambda_4 \mathcal{L}_{\text{normal}} + \lambda_5 \mathcal{L}_{\text{shared}}.\end{aligned} \tag{9}$$

We set $\lambda_1 = \lambda_2 = \lambda_4 = 10^{-3}$, $\lambda_5 = 10^{-4}$ and $\lambda_3 = 10^{-5}$, one to three orders of magnitude below the photometric term, and keep them fixed across all scenes and experiments.

4. Experiments

4.1. Setup

We use two datasets for our experiments: Mip-NeRF360 [2] and ScanNet++ [17]. We evaluate on five scenes from each, optimizing each for 30k iterations under an identical schedule, so the only variable between runs is which part of the method is enabled (see Tab. 2).

Metrics We evaluate our results by using PSNR, SSIM, and geometry against the reference mesh with Chamfer-L_1 distance and F-score. To evaluate on geometry consistency and semantic separation, we also add two mesh-intrinsic measures: normal consistency and the percentage of shared triangles, i.e. triangles whose three corner vertices do not all carry the same instance label.

4.2. Results on Mip-NeRF360

We present the results for Mip-NeRF360 [2] in Tab. 1, which also includes the ablation of the different regularization terms we introduce in this work. Because of computational constraints, we could not exactly reproduce the results given in the original MeshSplatting [5], since they run over all Mip-NeRF360 scenes and utilize super-resolution, which increases memory and time costs. Furthermore, Mip-NeRF360 does not provide semantic or geometric ground truth meshes, so we only run the first two ablations and report only purely photometric metrics (PSNR and SSIM). Our reproduction runs over a subset of 5 scenes, without super-resolution. We find that GeMS scores higher than the reproduced results on both metrics (24.62 vs. 23.10 dB PSNR and 0.743 vs. 0.651 SSIM), and the planar consistency losses keep being the strongest change proposed.

4.3. Results on ScanNet++

Tab. 2 reports the mean over the five scenes, and Fig. 2 shows a held-out view of one of them under every condition. Our first observation is that the weakness of the baseline is geometric rather than photometric: it renders held-out views within a decibel of every other condition, yet its Chamfer distance is close to twice that of any variant of our method. This is expected given the under-constrained nature of the photometric objective, with many mesh configurations explaining the training views equally well. Every component we add improves Chamfer distance and F-score without reducing photometric quality; normal consistency and triangle sharing, however, do not improve uniformly.

Table 1. Comparison on the Mip-NeRF360 dataset [2]. Our result is the mean over 5 scenes, while the original MeshSplatting is evaluated on the full test set. We report the **best** results for each experiment. Original results are excluded, as the test sets are different.

Method	**PSNR** ↑	**SSIM** ↑
MeshSplatting	24.78	0.728
MeshSplatting (reproduced)	23.10	0.651
MeshSplatting + $\mathcal{L}_{\text{geo}}$	23.14	0.665
MeshSplatting + $\mathcal{L}_{\text{planar}}$	24.46	0.739
GeMS (Ours)	**24.62**	**0.743**

Table 2. Evaluation on a subset of ScanNet++ [17], with an ablation isolating each of the introduced regularization losses. Losses are grouped as in Sec. 3: $\mathcal{L}_{\text{geo}}$ and $\mathcal{L}_{\text{planar}}$ each bundle two terms that are always enabled together. Best result is shown in **bold**.

	Initialization	PSNR↑	SSIM↑	Chamfer-L1↓	F-score↑	Normal Cons.↑	Shared Tri. %↓
MeshSplatting	SfM	19.625	0.781	0.110	0.540	0.692	12.33
MeshSplatting + $\mathcal{L}_{\text{geo}}$	PlanarSplatting	19.722	0.786	0.068	0.588	0.656	11.29
MeshSplatting + $\mathcal{L}_{\text{planar}}$	PlanarSplatting	**20.591**	0.792	**0.056**	0.629	**0.767**	12.85
MeshSplatting + $\mathcal{L}_{\text{shared}}$	PlanarSplatting	20.544	0.793	0.068	0.571	0.727	12.98
GeMS (Ours)	PlanarSplatting	19.995	**0.795**	0.062	**0.630**	0.762	**10.33**

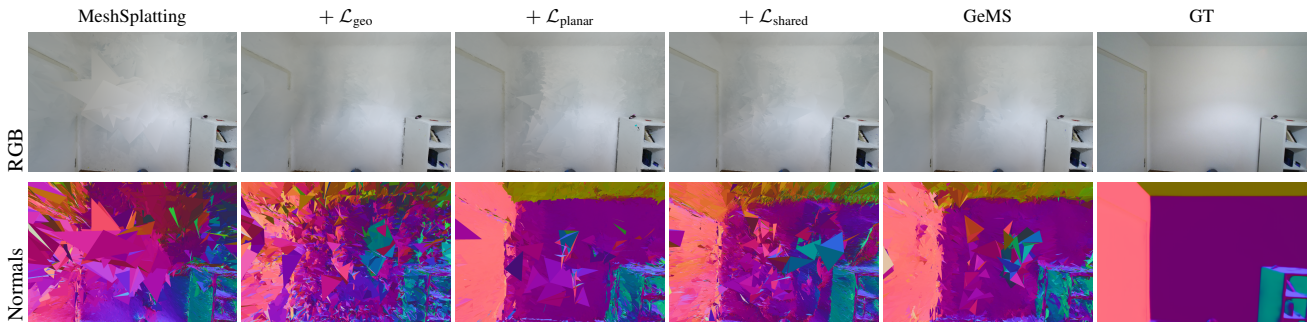


Figure 2. Qualitative comparison on a held-out view of a ScanNet++ scene. *Top:* rendered RGB. *Bottom:* rendered normal maps, which expose the underlying mesh geometry.

Plane alignment is the strongest single addition. Anchoring triangles to the detected planes gives the best result of any single term: the highest PSNR (20.591 dB), the lowest Chamfer distance (0.056) and the best normal consistency (0.767). Unlike the other two terms, it acts on both aspects of the geometry at once, fixing where the surface sits and how it is oriented, which is what the photometric objective leaves unconstrained. Its weak spot is on triangle sharing, where we report numbers above the baseline.

Smoothness alone cleans the surface without orienting it. The Laplacian and normal-consistency terms slightly lower the Chamfer distance and raise the F-score, but they barely move PSNR (19.722 vs. 19.625) and drive normal consistency below the baseline. While damping high-frequency vertex noise can remove the slivers the baseline accumulates over densification, it is hard to observe this effect in the presence of photometric optimization.

Semantic supervision needs a stable surface. In isolation, the semantic-boundary term is the one result that runs against our expectations. While it improves appearance and geometry, it makes the shared triangles quantity slightly worse than the baseline (12.98% vs. 12.33%), yet the same term drives that quantity to its lowest value of any condition once the other regularizers are active (10.33%). We attribute this behavior to high object proximity in the evaluated scenes, which makes the process of generation of new triangles challenging. When coupled with geometry regularization losses (which reduce the number of overall triangles), this problem is partially alleviated, allowing our method to finally improve the shared triangles metric.

5. Conclusion

We introduce multiple regularization and initialization techniques based on geometric priors to the MeshSplatting pipeline. Our results show that a previous decomposition into a small set of large planes covering the most significant areas of the scene helps significantly with the fidelity of the reconstruction, both geometric and appearance-wise.

However, our method still has some limitations. We observe in our experiments that the semantic separation loss introduced in Sec. 3.3 fails to decrease the number of shared triangles on ScanNet++. Also, the generated geometry still has room for improvement (cf. the normal maps in Fig. 2). As future work, we devise an adaptive triangle creation scheme to alleviate the shared semantic information and incorporate stronger regularization terms to improve the geometry of the surfaces.

References

[1] Elena Alegret, Kunyi Li, Sen Wang, Siyun Liang, Michael Niemeyer, Stefano Gasperini, Nassir Navab, and Federico Tombari. Gala: Guided attention with language alignment for open vocabulary gaussian splatting. In *International Conference on 3D Vision (3DV)*, 2026. 1

[2] Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. Mip-NeRF 360: Unbounded anti-aliased neural radiance fields. In *CVPR*, pages 5470–5479, 2022. 3

[3] Antoine Guédon and Vincent Lepetit. SuGaR: Surface-aligned gaussian splatting for efficient 3d mesh reconstruction and high-quality mesh rendering. In *CVPR*, pages 5354–5363, 2024. 1

[4] Jun Guo, Xiaojian Ma, Yue Fan, Huaping Liu, and Qing Li. Semantic gaussians: Open-vocabulary scene understanding with 3d gaussian splatting. *IEEE TCSVT*, 36(7):9265–9277, 2026. 2

[5] Jan Held, Sanghyun Son, Renaud Vandeghen, Daniel Rebain, Matheus Gadelha, Yi Zhou, Anthony Cioppa, Ming C. Lin, Marc Van Droogenbroeck, and Andrea Tagliasacchi. MeshSplatting: Differentiable rendering with opaque meshes. In *CVPR*, pages 7320–7329, 2026. 1, 2, 3

[6] Jan Held, Renaud Vandeghen, Adrien Deliege, Abdullah Hamdi, Anthony Cioppa, Silvio Giancola, Andrea Vedaldi, Bernard Ghanem, Andrea Tagliasacchi, and Marc Van Droogenbroeck. Triangle splatting for real-time radiance field rendering. In *International Conference on 3D Vision (3DV)*, 2026. 1

[7] Binbin Huang, Zehao Yu, Anpei Chen, Andreas Geiger, and Shenghua Gao. 2d gaussian splatting for geometrically accurate radiance fields. In *ACM SIGGRAPH Conf. Papers*, 2024. 1

[8] Shahram Izadi, Richard A. Newcombe, David Kim, Otmar Hilliges, David Molyneaux, Steve Hodges, Pushmeet Kohli, Jamie Shotton, Andrew J. Davison, and Andrew Fitzgibbon. Kinectfusion: Real-time dynamic 3d surface reconstruction and interaction. In *ACM SIGGRAPH 2011 Talks*, pages 23:1–23:1, New York, NY, USA, 2011. ACM. 1

[9] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM TOG*, 42(4), 2023. 1

[10] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020. 1

[11] Songyou Peng, Michael Niemeyer, Lars Mescheder, Marc Pollefeys, and Andreas Geiger. Convolutional occupancy networks. In *ECCV*. Springer, 2020. 1

[12] Giovanni Pintore, Claudio Mura, Fabio Ganovelli, Lizeth Fuentes-Perez, Renato Pajarola, and Enrico Gobbetti. State-of-the-art in automatic 3d reconstruction of structured indoor environments. *Comput. Graph. Forum*, 39(2):667–699, 2020. 1

[13] Yawar Siddiqui, Lorenzo Porzi, Samuel Rota Bulò, Norman Müller, Matthias Nießner, Angela Dai, and Peter Kontschieder. Panoptic lifting for 3D scene understanding with neural fields. In *CVPR*, pages 9043–9052, 2023. 1

[14] Bin Tan, Rui Yu, Yujun Shen, and Nan Xue. PlanarSplatting: Accurate planar surface reconstruction in 3 minutes. In *CVPR*, pages 1190–1199, 2025. 2

[15] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. In *NeurIPS*, 2021. 1

[16] Mingqiao Ye, Martin Danelljan, Fisher Yu, and Lei Ke. Gaussian grouping: Segment and edit anything in 3d scenes. In *ECCV*, pages 162–179, 2024. 2

[17] Chandan Yeshwanth, Yueh-Cheng Liu, Matthias Nießner, and Angela Dai. ScanNet++: A high-fidelity dataset of 3d indoor scenes. In *ICCV*, pages 12–22, 2023. 3, 4